Hadi Malik

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Education

Georgia Institute of Technology Atlanta, GA
B.S. Computer Science | GPA: 4.0 *Expected: December 2026*
- Relevant Coursework: Data Structures, Object-Oriented Programming, Machine Learning, Information Security, System Design, Linear Algebra

Experience

Incoming Software Engineer Intern June 2025
Genuine Parts Company *Atlanta, GA*

Undergraduate Researcher August 2025 – Present
Georgia Tech School of Cybersecurity and Privacy *Atlanta, GA*
- Quantified privacy risks in **Convolutional Neural Networks (CNN)** by **measuring data-leakage** in inference outputs on a 60,000+ image dataset.
- Developed a **statistical evaluation framework** to assess model robustness, utilizing **probabilistic analysis** to detect AI privacy vulnerabilities.
- Engineered a **Machine Learning pipeline** for end-to-end model training and inference, enabling scalable and reproducible experiments.

Software Engineer Intern May 2025 – July 2025
In Good Company Works *Atlanta, GA*
- Engineered a full-stack AI platform using **React, Node.js, and Firebase**, building a **Retrieval-Augmented Generation (RAG)** pipeline for context-aware responses; **optimized LLM accuracy** through iterative feedback loops to validate usability across **50+ sessions with 15 clients.**
- Created **high-throughput RESTful APIs** on **Google Cloud (GCP)**, optimizing data serialization to **reduce backend latency by 50%** for scalability.
- Hardened API security posture by implementing **RBAC, JWT authentication, CORS compliance,** and strict session management protocols.
- Launched native **iOS companion app** using SwiftUI and MVVM, achieving real-time data sync with the web application via **shared API endpoints.**

Software Engineer Intern May 2024 – August 2024
Verifacto Inc. *Sandy Springs, GA*
- Achieved **99% model accuracy** for document classification by training a **Reinforcement Learning** agent on **10,000+ insurance documents**.
- **Improved query efficiency by 40%** by migrating **MySQL** to **vectorized MongoDB**, optimizing retrieval for high-dimensional data.
- Engineered a LangChain interface that enables clients to **query databases using Natural Language** (NLP), to generate **actionable insights.**
- **Increased user engagement by 15%** by implementing internationalization (i18n) pipelines to localize content for Spanish-speaking clients.

Data Science Research Assistant January 2025 – Present
Georgia Tech VIP - ACT Driving Simulator *Atlanta, GA*
- **Achieved 95% accuracy** in detecting stress by applying **CUSUM anomaly detection** algorithms to **EEG time-series data** from driver simulations.
- Optimized feature extraction by using **Welch's Method** and **Power Spectral Density (PSD)** to isolate key spectral features (Alpha, Beta, Gamma).
- Built a MATLAB preprocessing pipeline to clean raw data using z-score normalization, ensuring high data quality for statistical modeling.

Projects

Urology Report Generator | Python, PyTorch, OpenCV, LLMs, LoRA, Linux/Bash, Slurm December 2025 - Present
- Fine-tuned MedGemma LMM using LoRA on **600+ studies**, achieving **94.8% semantic similarity** to facilitate automated clinical documentation.
- Optimized GPU memory via Flash-Attention, enabling a **5x increase in image processing capacity** (from 20 to 100+ images per study).
- **Accelerated automated report generation to <60 seconds** per report by deploying a high-performance pipeline on NVIDIA H200 GPUs via Slurm.
- Developed an **evaluation pipeline** with BiomedCLIP embeddings to **quantify performance and reduce hallucinations** in clinical environment.

Credit Card Fraud Detector | Python, scikit-learn, PyTorch, Numpy, Pandas, Matplotlib, Deep Learning September 2025 - December 2025
- Developed a **Neural Network architecture** in PyTorch to detect credit card fraud, **achieving 0.90 ROC-AUC** on a dataset with 590K transactions.
- Engineered an **ML pipeline** using **scikit-learn** and **Pandas**, implementing **PCA dimensionality reduction** to optimize the high-dimensional data.
- **Optimized performance** by analyzing precision-recall curves to tune thresholds, **minimizing false negatives** in real-world transaction scenarios.

Network Vulnerability Scanner | Python, Docker, Socket Programming, TCP/IP, REST APIs, Multi-threading June 2025 - July 2025
- Built multi-threaded Python vulnerability scanner for open port detection, banner grabbing, and CVE mapping via NVD CPE API.
- Implemented CPE-based CVE matching with CVSS filtering, CLI support, and interactive subnet host discovery for vulnerability assessment.
- Generated HTML/JSON reports and tested scanner accuracy against Dockerized vulnerable services (vsFTPd, OpenSSH, MailHog, Apache).

Skills

- **Languages**: Python, Java, C, R, JavaScript, SQL, Swift, Assembly (x86), PHP, TypeScript, Go, Zsh
- **AI/ML**: PyTorch, NumPy, Pandas, Keras, Langchain, ChatGPT, Google Gemini
- **Web Development**: React, Django, Flask, Firebase, MongoDB, MySQL, WordPress, Webflow
- **Tools and Cloud**: Git, Docker, AWS, Google Cloud, Matlab, Wireshark, MySQL, Salesforce